

Software & Algorithm

Arduino IDE:

- The software used to program the board is
- It is used to write, compile, and upload code to the Arduino Uno.

Arduino Uno:

- It is the main controller of the system
- Executes the program written in Arduino IDE
- Controls input and output devices
- It controls the ultrasonic sensor through programmed instructions

It sends signals to the ultrasonic sensor, processes the received data, and controls the buzzer.

System Configuration:

1. Define Constants: TRIG_PIN : Digital Output

- ECHO_PIN : Digital Input
- BUZZER_PIN : Digital Output
- THRESHOLD : 25cm
- V_SOUND : 0.0343cm per microseconds (Speed of sound at sea level)

Phase 2: Signal Generation (The Trigger)

2. Stabilize: Set TRIG_PIN to LOW for 2microseconds ensure a clear pulse.
3. Initiate Pulse: Set TRIG_PIN to HIGH for exactly 10microseconds.
4. Terminate Pulse: Set TRIG_PIN to LOW.

Phase 3: Data Acquisition & Calculation

5. Listen: Use the microcontroller to measure the duration (t) the ECHO_PIN remains HIGH.
6. Compute Distance: Calculate distance using the time-of-flight formula:

Phase 4: Logic Execution

7. Output Data: Transmit the value of t to the Serial Monitor for debugging.
8. Evaluate Condition:
 - * IF distance $\leq$ 25 cm Turn buzzer ON (Alert user).
 - * ELSE: Set BUZZER_PIN to LOW (Safe distance).

Phase 5: Iteration

9. Cooldown: Pause execution for 1000ms to allow sound reflections to dissipate.

10. Repeat: Return to Phase 2.

CODE:

```
long duration;
int distance;

const int TRIG = 9;
const int ECHO = 10;

int buzzer = 8;

void setup() {
  pinMode(TRIG, OUTPUT);
  pinMode(ECHO, INPUT);
  Serial.begin(9600);

  pinMode(buzzer, OUTPUT);
}

void loop() {
  // Clear trig pin
  digitalWrite(TRIG, LOW);
  delayMicroseconds(2);

  // Send 10us pulse
  digitalWrite(TRIG, HIGH);
  delayMicroseconds(10);
  digitalWrite(TRIG, LOW);

  // Read echo
  duration = pulseIn(ECHO, HIGH);
```

```
// Convert to distance (cm)
distance = duration * 0.034 / 2;

// Print distance
Serial.print("Distance: ");
Serial.print(distance);
Serial.println(" cm");

// Activate buzzer if distance <= 25 cm
if (distance <= 25) {
    digitalWrite(buzzer, HIGH); // Turn buzzer ON
    delay(50);
    digitalWrite(buzzer, LOW);
} else {
    digitalWrite(buzzer, LOW); // Turn buzzer OFF
}

delay(500);
delay(500);
}
```